Martin Gregory-Eaves

Munich, Germany | martin.gregory-eaves@tum.de | +48 500 484 111 | martingecodes.github.io | linkedin.com/in/martin-ge

Profile

MSc MRBE at TUM with a engineering research background spanning embedded systems, control theory, and aerodynamics. My interests in swarm control of miniaturized robots and flight control stem from two the research projects I was a part of and amounted to 25 ECTS of credits. My previous training comes from a applied physics BSc from which I gained fundamental knowledge in linear algebra, real analysis, signal processing and thermodynamics.

Education

Oct 2026 – Present | Munich, Germany

MSC MECHATRONICS, ROBOTICS & BIOMECHANICAL ENGINEERING (MRBE) Technical University of Munich

- Key courses: Machine Learning & Optimization, Robotics, Advanced Programming in C++, Autonomous Vehicles: Development and Simulation, Advanced Control, Embedded Systems & Security, Computational Neuroscience.

Sep 2023 – Jun 2026 | Groningen, Netherlands

BSC APPLIED PHYSICS University of Groningen

- Extra course (5 ECTS): Mechatronics.
- Thesis (15 ECTS): "An Investigation into Gust Mitigation Strategies for Small Fixed-Wing Aerial Vehicles" - Lentink Lab.
- Research project (10 ECTS): "Experimental Modal Analysis in a Swarm of Miniaturized Vibration-Sensing Robots" - DAISY Lab.
- **Leadership:** Led the Cybathlon thumb actuation team. Tutored IB math and physics 10 hours/ week. Raised £350 for charity running the London Landmarks Half Marathon

Work Experience

Mar 2025 – Sep 2025 | Groningen, Netherlands

UNDERGRADUATE RESEARCH ASSISTANT DAISY Lab

- Automated FEM modal analysis of sheet-metal panels in Abaqus using Python scripts, comparing vibration response across sensor configurations and structural defects.
- Designed and 3D-printed a robot enclosure and custom sensor mount in Fusion 360 for experimental vibration testing.
- Processed experimental data in Python using FFT and power spectral density analysis to validate simulation results.
- Reviewed literature on kriging interpolation as a search algorithm for defect localisation.

Jul 2025 – Aug 2025 | Vancouver, Canada

MECHATRONICS ENGINEERING INTERN Aspect Biosystems

- Used CAD to redesign and improve alignment accuracy of silicone sheet bonding apparatus.
- Created engineering drawings with accurate GD&T approved by in-house machine shop.
- Collaborated on various engineering assemblies and parts through Solidworks PDM.

Nov 2024 – Mar 2025 | Groningen, Netherlands

THUMB ACTUATION ENGINEER Cybathlon

- Designed a component of the thumb actuation system using CAD to enable reliable grasping of objects with varying geometries.
- Integrated Hall-effect sensors into the thumb joint to enable real-time joint angle measurement and support object detection.
- Selected as one of six on-site engineers to provide technical support at the ETH Cybathlon 2024.

Apr 2026 – Jun 2026 | Groningen, Netherlands

UNIVERSITY TEACHING ASSISTANT University of Groningen

- Oversaw groups of 4-5 students participating in their 1st year lab design course.
- Provided constructive feedback in in-person meetings and graded draft and final lab reports.
- Gave advice on project management and self-organization to younger students.

Projects

2026 | University of Groningen

BIRD-INSPIRED PRESSURE-SENSING WING BSc Thesis, Lentink Lab

- Embedded 11 BMP280 pressure sensors in a 3D-printed Eppler 423 wing to detect incoming gusts.
- Wrote C++ firmware in the Arduino IDE for a Teensy 4.1, reading all sensors over SPI.
- Wind-tunnel tested under pulsing and shear turbulence; distinguished steady from turbulent flow.

- **Tools:** C++, Arduino IDE, Teensy 4.1, SPI, BMP280, SolidWorks, KiCad.

LINE-FOLLOWING ROVER

- Coded a PID algorithm so a pre-built RC car follows a black line and avoids obstacles.
- **Tools:** PID control, infrared and sonar sensors.

Skills

Programming	C++, Python, C, Office 365, Matplotlib, Numpy, Pandas
Hardware	Teensy 4.1, Arduino IDE, SPI / I^2C, pressure and Hall-effect sensors, soldering
CAD & Simulation	SolidWorks, Fusion 360, Abaqus, KiCad, 3D printing
Languages	English (native), Polish (native), Spanish (C1–C2)

Interests & Hobbies

I am a big FC Barcelona fan and grew up playing football in school and at university. Recently I have taken up long distance running and bouldering. I also enjoy playing the guitar and reading both fiction and non-fiction.